Wildfire-Diff: A Controllable Diffusion Framework for Synchronized Generation of Remote Sensing Wildfire Images and Masks

Junyi Wu, Libo Wang, Xiaoliang Meng, Haorong Liang, Xiangyu Chen, Xiaokang Zhang, and Guisong Xia

***Abstract*—Burned area mapping based on remote sensing images is a key task in wildfire monitoring and holds significant importance for disaster assessment. In recent years, deep learning (DL) methods have made significant progress in this task and are gradually becoming the mainstream approach. However, the issue of existing training sample scarcity severely limits the accuracy and generalization of DL-based burned area mapping methods. To address this issue, we propose a controllable diffusion model, namely Wildfire-Diff, for synchronized generation of paired wildfire images and pixel-level masks. Different from current conditional diffusion models, we strengthen the realness of generated samples by constraining the spatial distribution and background of wildfires. Specifically, we design a vegetation-constrained wildfire mask generation network (VCMGNet) that can restrict burned areas to vegetation-covered regions, thereby reducing unrealistic fire scenes (e.g., wildfire on the river). Furthermore, we develop a background-aware wildfire image generation network (BAIGNet) to reduce the randomness of the generated image background and mitigate the distortion issue. Benefiting from these two frameworks, our Wildfire-Diff can generate high-quality wildfire change-detection (CD) samples to promote burned area mapping. Experimental results demonstrate the superiority of our Wildfire-Diff (30.42 FID) compared to other advanced diffusion models (e.g., ControlNet). Meanwhile, in downstream burned area mapping tasks, applying wildfire samples generated by our Wildfire-Diff can significantly improve the accuracy intersection over union (IoU) of four CD models (i.e., BIT, CGNet, EfficientCD, and BAMCD) by at least 20%, which further reveals its great potential to address training sample scarcity and promotes downstream applications.**

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***Index Terms*—Burned area mapping, change detection (CD), diffusion model, trainable sample generation.**

I. Introduction

WILDFIRE is a key disturbance factor in ecosystems and occurs widely in a variety of forest scenes, from tropical rainforests to boreal coniferous forests. In recent years, the frequency, scale, intensity, and breadth of wildfires have all shown a significant upward trend, causing serious damage to the regional and even global social economy and ecological balance. There is an urgent need to accurately extract the burned area map, thus understanding the spatiotemporal dynamics and evolution of wildfires. Driven by this, burned area mapping based on remote sensing images is becoming a hot research topic.

Traditional burned area mapping methods usually mine the band response, vegetation coverage, moisture content, and other physical characteristics of the burned area and design remote sensing spectral indexes (e.g., NBR and BAI) for extraction. These methods are computationally simple and interpretable but demonstrate limited accuracy due to their restricted ability to capture high-level semantic information in remote sensing images. With the rapid development of artificial intelligence (AI), data-driven deep learning (DL) methods, like change detection (CD) [1], [2], [3], offer a more promising solution for remote sensing the mapping of burned area. For example, Knopp et al. [4] designed a UNet-like semantic segmentation network to significantly improve the extraction accuracy of European burned areas. Sui et al. [5] introduced the self-attention mechanism to construct a twin CD model for burned area mapping, which performed well in various wildfire scenes compared with the traditional methods. The performance of DL methods, although encouraging, relies on training with large-scale and high-quality annotated data. Limited training samples severely restrict the accuracy and generalization of DL-based burned area mapping methods. However, as for wildfire remote sensing images, it is very difficult to obtain a mass of finely labeled masks due to the complex and fragmented spatial distribution of burned areas. Thus, how to address the annotated data scarcity has become the core issue in this field.

Recently, generative AI technology has attracted much attention in computer vision and provides a new solution for addressing the annotated data scarcity issue. For example, Zheng et al. [6] proposed a scalable multitemporal method

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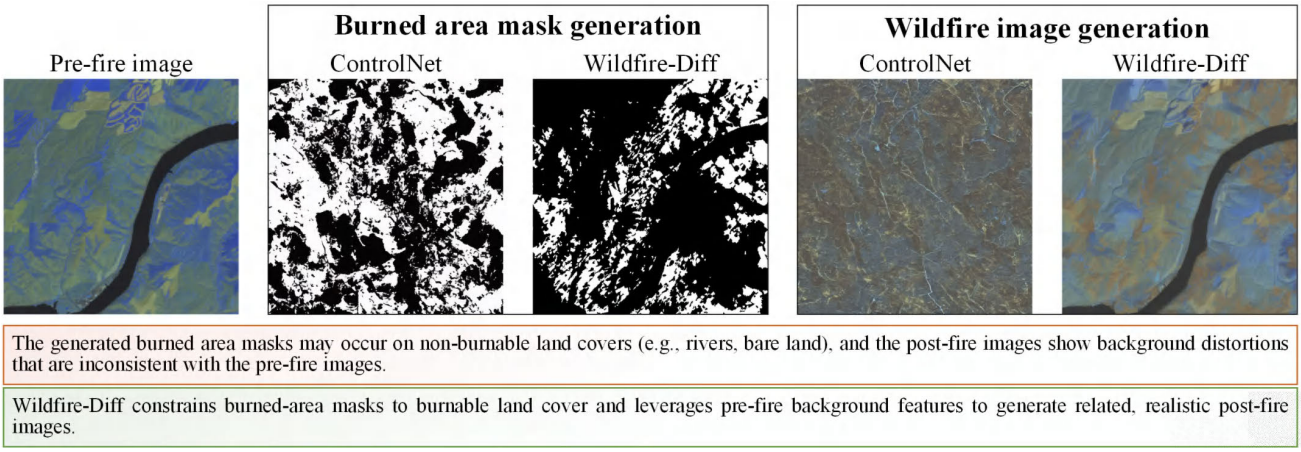


Fig. 1. Comparison between ControlNet and our wildfire-diff in synchronized generation of wildfire images and masks.

for generating remote sensing images, called Changen, which can automatically generate trainable CD samples through two GAN-driven modules. Zang et al. [7] introduced a diffusion-model-based data generator for CD, namely ChangeDiff, which leverages a multiclass distribution-guided text prompt and semantic layouts to generate massive CD samples for improving the performance of the model. Especially, the milestone ControlNet introduces extra multimodal conditional signals (e.g., text, image, and metadata) based on the foundational diffusion model, stable diffusion (SD), to control the details of generated images (e.g., color, style, and layout), greatly enhancing the realness and quality of generated images as well as the efficiency of generation. However, current controllable image generation methods still face two limitations in wildfire CD samples generation as follows.

1) Random spatial distribution of burned area. For instance, as shown in Fig. 1, ControlNet could generate burned areas in error locations (e.g., on the bare land or rivers) due to the lack of a spatial-constraint mechanism, thereby affecting the realism of generated wildfire images.
2) Background distortion of postfire images. The background of bitemporal wildfire remote sensing images demonstrates high similarity and relevance. However, the ControlNet could generate unrelated background to prefire images due to the lack of a background-controllable mechanism.

In this article, we propose a novel diffusion-based image controllable generation framework, namely Wildfire-Diff, for synchronized generation of high-quality wildfire images and masks. Specifically, our Wildfire-Diff is a two-stage wildfire CD sample generation framework, which is of great potential in addressing the trainable sample scarcity of burned area mapping. First, we develop a vegetation-constrained wildfire mask generation network (VCMGNet) to ensure the generated burned areas are in reasonable locations. Second, we further design a background-aware wildfire image generation network (BAIGNet) to strengthen the realness of the generated postfire background, thereby improving the applicability of generated wildfire CD samples in burned area mapping. The main contributions of this work are as follows.

1) We propose Wildfire-Diff, a novel two-stage diffusion-based controllable generation framework for synthesizing high-quality paired wildfire images and binary masks. Compared with baseline methods, Wildfire-Diff achieved an average relative improvement of 92.64% in intersection over union (IoU).
2) To improve the controllability and realism of wildfire generation, we design a VCMGNet and a BAIGNet. The former leverages vegetation indices to reduce the random spatial distribution of generated burned areas, while the latter incorporates prefire images to strengthen correspondence and alleviate background distortion of postfire images.
3) Extensive experiments on multiple wildfire CD models demonstrate that Wildfire-Diff can serve as an effective data augmentation tool in data-scarce burned area mapping scenarios. By complementing real samples with diverse synthetic wildfire patterns, it consistently improves downstream detection performance and enhances the practical value of wildfire image generation for remote sensing applications.

II. Related Work

A. Generation of Remote Sensing Images

In recent years, AI-generated content (AIGC) has advanced rapidly. By using large language models (LLMs), generative adversarial networks (GANs), diffusion models, and vision–language models (VLMs), researchers can produce text, images, and other media from diverse inputs. In computer vision, diffusion techniques have overtaken both GANs [8], [9] and autoregressive transformers [10], [11] to become the state of the art in image generation and editing [12], [13], [14], [15]. By combining diffusion with contrastive VLMs such as CLIP [9], text-to-image systems like Imagen [15], DALL-E2 [13], GLIDE [12], and the SD [14], these models excel not only in aligning vision and language but also in raising the bar for image realism, diversity, and controllability. Although they have transformed natural-scene generation, their application to specialized areas, such as remote sensing images, remains largely unexplored.

Despite success in natural imagery, applications in specialized fields, particularly remote sensing, remain limited. Early diffusion-based remote sensing generators have shown

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promise. For example, RSDiff [16] adopts the text-conditioned generation, implementing a two-stage diffusion architecture that enhances detail reconstruction capabilities by progressively refining image resolution. CRS-Diff [17] has realized controllable generation of high-resolution remote sensing images under the ControlNet [18] framework. Word2Scene leverages a diffusion model to generate complex scenes from a single word. This method incorporates domain-specific geographic knowledge to optimize remote sensing scene descriptions, thereby enhancing the accuracy and interpretability of the input prompts [19]. DiffusionSat [20] innovatively integrates multimodal inputs, combining numerical metadata with textual descriptions within the SD framework, significantly improving the physical consistency of generated images. The recently proposed SyntheWorld [21] methodology synergizes 3-D modeling with LLMs and SD for automated synthetic data production in cartography and CD applications. However, its potential value for downstream remote sensing tasks, such as classification and object detection, remains underexplored.

To alleviate the scarcity of annotated RS data, existing work relies on augmentation (e.g., copy–paste pipelines for buildings) [22] or mask-based semantic stitching with probabilistic graphical models to simulate temporal change [6]. Newer studies explore diffusion for semantic-aware, multitemporal change datasets [7]. But most require abundant samples and struggle in data-poor scenarios like wildfire monitoring, where domain adaptation and fine-grained control are critical yet underdeveloped.

B. Burned Area Mapping

Traditional burned area detection methods generally employ remote sensing spectral indices. By devising band ratios sensitive to vegetation loss or combustion characteristics [e.g., normalized burn ratio (NBR), burned area index (BAI), and mid-infrared burn index (MIRBI)], these approaches delineate fire-affected regions. Index-based techniques offer high interpretability and computational efficiency and underpin widely used global products, such as MCD64A1 and FireCCI51. However, they depend heavily on expert knowledge and often fall short of capturing intricate spatial patterns in remote sensing images.

DL has renewed burned area mapping [23] methods by applying semantic segmentation [24], [25], [26] and CD models [27], [28], [29]. U-Net variants demonstrated significant improvements in extraction accuracy, achieving up to 90% precision when processing Sentinel-2 multispectral imagery for typical European fire regions [4]. Furthermore, U-Net architectures incorporating multilevel feature fusion outperformed traditional index-based algorithms by over 15% on average across five major global fire zones [30]. Advanced implementations utilizing self-attention mechanisms in "Siamese" network configurations enabled parallel analysis of prefire and postfire images, thereby enhancing the robustness of CD processes [5]. However, the scarcity of annotated data remains a major challenge for these approaches. Research indicates that DL models trained and validated on such limited datasets often exhibit poor generalization across different scenarios, leading to suboptimal accuracy levels. This shortfall hinders the operational use of these models for producing reliable burned area extraction products. Furthermore, available burned area datasets are primarily composed of remote sensing images from geographically restricted regions. These datasets suffer from insufficient sample sizes lack of data diversity and fail to account for the spatiotemporal variability of burned area characteristics across different environmental contexts, such as varying climate zones and topographical features. Although DL methods have demonstrated clear advantages over traditional index-based approaches, further research is needed to refine feature modeling and enhance generation across multiple scenarios. Such progress depends on the availability of large-scale, high-quality, and multiscenario datasets.

III. Methods

To effectively address the above issues and challenges, we propose the Wildfire-Diff framework for generating paired postfire remote sensing images and masks in a controllable manner. As illustrated in Fig. 2, the proposed Wildfire-Diff is composed of the VCMGNet and the BAIGNet. Sections III-A–III-C provide a detailed description of each module.

A. Overview of the Wildfire-Diff

Built upon the latent diffusion model (LDM) [14], which serves as the base generator due to its strong image synthesis capability, Wildfire-Diff incorporates both VCMGNet and BAIGNet as adapted variants of LDM. As illustrated in Fig. 2, the framework accepts three inputs: a prefire image x_p, text prompts y_1 and y_2, and a semantic layout of burned area coverage α. These inputs are fed into VCMGNet, which produces a binary burned area mask. This mask then serves as input to the downstream BAIGNet, ultimately generating the final output, the postfire remote sensing image.

In LDM, noise sampled from a normal distribution ϵ is added to the latent representation of an image over t steps. A U-Net [31] is then trained to predict the added noise distribution ϵ_θ. By reversing this diffusion process, the model iteratively denoises the latent representation to reconstruct the image. The training objective is defined as follows:

$$L_{LDM} = E_{\ell_0,t,\epsilon}\|\epsilon - \epsilon_\theta(\ell_t, t, \tau_\theta(P))\|^2 \quad (1)$$

where $\ell_0 = \epsilon_\theta(x)$ denotes the latent encoding of the image, and τ_θ is a pretrained text encoder [9] that incorporates cross-attention between the text prompt P and the latent state ℓ_t to guide the generation.

B. Vegetation-Constrained Wildfire Mask Generation Network

As shown in Fig. 3, VCMGNet consists of a burned area mask diffusion module and a vegetation-constrained block. It uses prefire images to generate the burned area mask at appropriate locations, thus reducing the randomness in spatial distribution. Specifically, the process consists of two main components: first, a burned area mask M_{syn} is synthesized by the diffusion module conditioned on the text prompt y.

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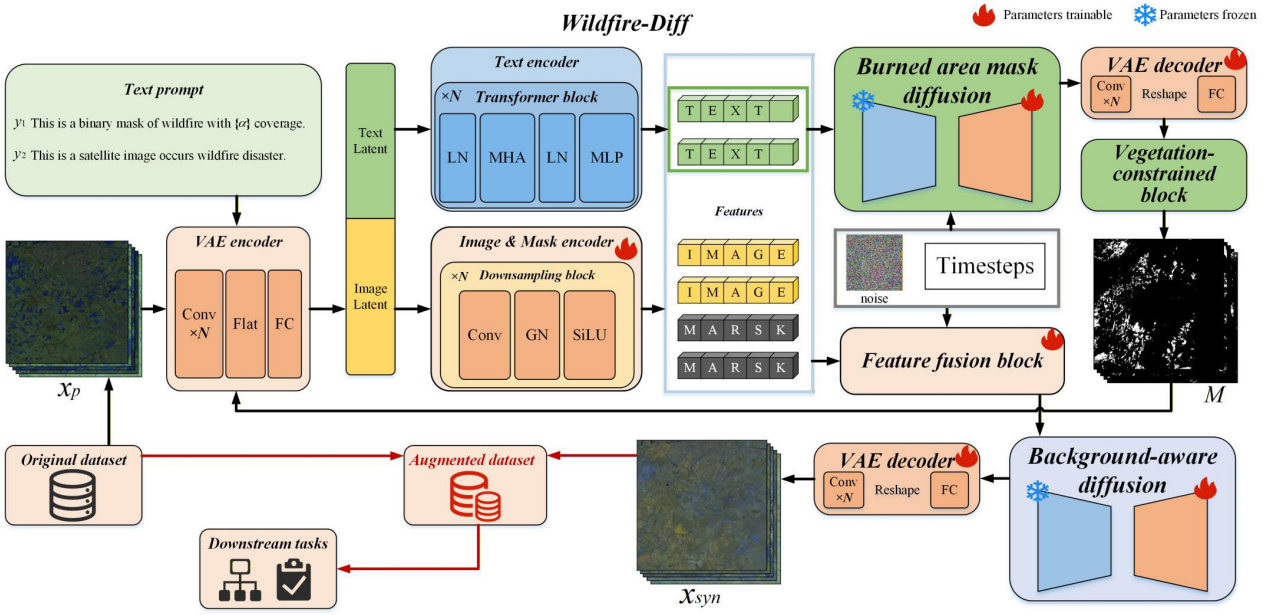


Fig. 2. Overview of Wildfire-Diff. y is text prompt, x_p is prefire image, α is burned area coverage prompt, M_{syn} is generated burned area mask, M_V is NDVI constraint mask, M is vegetation-constrained generated burned area mask, and x_{syn} is generated wildfire result. In this process, the original dataset is input to the system, then augmented with the model's output to create an augmented dataset. This enhances the performance of downstream tasks.

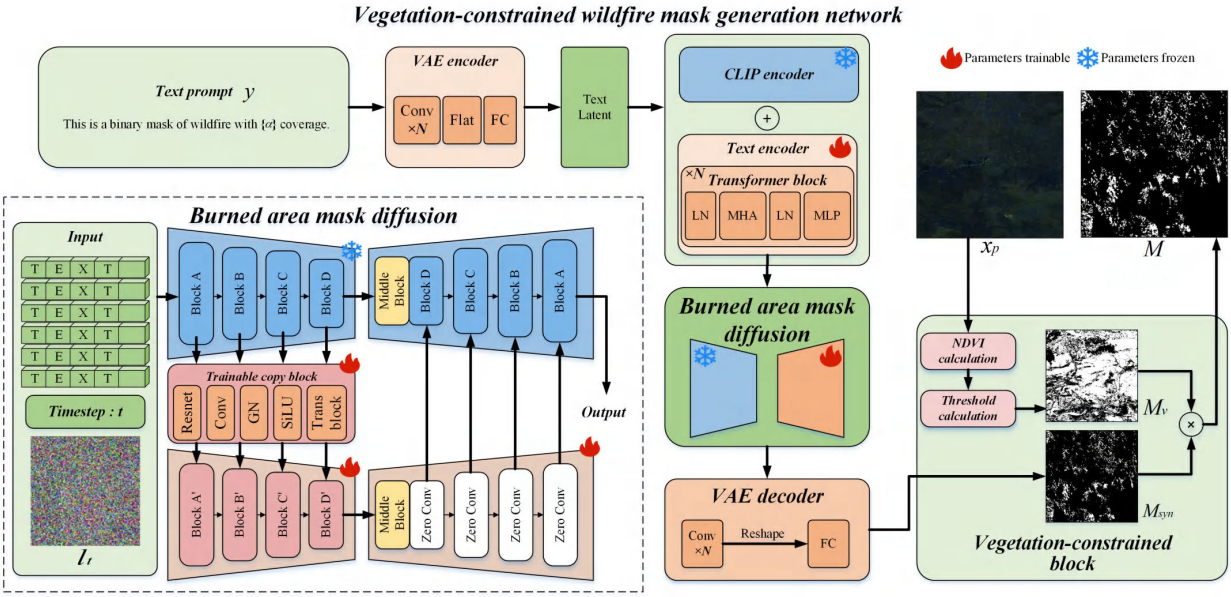


Fig. 3. Structure of VCMGNet, x_p is prefire image, y is text prompt, α is burned area coverage prompt, ℓ_t is the encoding of the image in the latent space, M_V is NDVI constraint mask, M_{syn} is generated burned area mask, and M is vegetation-constrained generated burned area mask.

Then, a vegetation constraint mask M_v is derived from the prefire image using the normalized difference vegetation index (NDVI; values >0.1 set to 1, else 0). The final mask M is obtained via elementwise multiplication

$$M = M_v \cdot M_{syn}. \tag{2}$$

This constraint limits burned areas to vegetated regions, avoiding spurious detection over nonvegetated surfaces, such as water, bare soil, or clouds. We used NDVI > 0.1 to define vegetated areas because very low NDVI values (0.1 or less) typically represent barren surfaces, whereas positive NDVI values above this level indicate vegetation, and NDVI threshold methods have been applied in previous vegetation classification studies [32], [33].

The burned area mask diffusion module is based on a fine-tuned LDM that generates binary masks conforming to the morphology of burned areas. Text prompts containing information about burned area coverage α are encoded using a customized CLIP text encoder $\mathcal{M}_{\tau\theta}$. This enables explicit textual control over the spatial extent of fire scars as follows:

$$\epsilon_\delta = \epsilon_\theta\left(\ell_t, t, \mathcal{M}_{\tau\theta}(y+\alpha)\right) + \epsilon_c\left(\ell_t, t, \mathcal{M}_{\tau\theta}(y+\alpha)\right) \tag{3}$$

$$\mathcal{L}_{mask} = \mathbb{E}_{\ell_0,t,\epsilon}\|\epsilon - \epsilon_\delta\|^2 \tag{4}$$

where ϵ_δ is the noise distribution prediction in ControlNet. $\mathcal{M}_{\tau\theta}$ is the pretrained text encoder our modified. ℓ is the

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Algorithm 1 Pseudocode for VCMGNet

Require: x_p: prefire image
Require: y_α: text prompt for burned area mask generation
Require: model: diffusion model for generating mask
Require: t: time step
Ensure: m: final burned area mask
  function VCMGNET(x_p, y_α, model, t)
    $\text{embeds}_{text} \leftarrow \text{encoder}(y_\alpha)$
    $l \leftarrow \text{model.initlatent}(\text{embeds}_{text})$
    $\text{noise} \leftarrow \text{model.step}(t, l, \text{embeds}_{text})$
    $\text{pred}_{sd} \leftarrow \text{model.unet}(\text{noise}, t)$
    $\text{pred}_{ctrl} \leftarrow \text{model.unet}(\text{noise}, t)$
    $\text{pred} \leftarrow \text{pred}_{sd} + \text{pred}_{ctrl}$
    $m_{\text{syn}} \leftarrow \text{model.decode}(\text{pred})$
    $\text{nir} \leftarrow x_p[\text{NIR}]$
    $\text{red} \leftarrow x_p[\text{Red}]$
    $\text{ndvi} \leftarrow (\text{nir} - \text{red})/(\text{nir} + \text{red} + 10^{-6})$
    $m_v \leftarrow (\text{ndvi} > 0.1)$ converted to integer
    $m \leftarrow m_v \times m_{\text{syn}}$
    return m
  end function

encoding of the features in the latent state. Pseudocode for the full procedure is provided in Algorithm 1.

C. Background-Aware Wildfire Image Generation Network

To generate spectral features of wildfire within the burned area while simultaneously preserving the features of the background (unburned) regions, thereby mitigating background distortion. As illustrated in Fig. 4, the prefire image is incorporated during both fine-tuning and generation to provide background context. The conditioning signal for the diffusion model is composed as

$$\ell_{BAIGNet} = \ell_t + \ell_{condition}(M) + \ell_{background}(x_p) \tag{5}$$

where ℓ_t is the latent-space feature at generation step t, $\ell_{condition}(M)$ is the latent embedding of the burned mask, and $\ell_{background}(x_p)$ represents the prefire image encoded into the latent space.

An inpainting mechanism is introduced during inference to further improve background preservation. The generated image x_{syn} and prefire image x_p are cropped using the burned area mask M and its complement M_c, respectively. Both regions are noised, merged, and denoised to produce the final output

$$x_i = Denoise\left(M \cdot x_{syn} + Addnoise\left(M_c \cdot x_p\right)\right). \tag{6}$$

Although this strategy enhances visual consistency, it may limit performance in downstream CD tasks. Therefore, the inpainting block is only included in ablation studies and omitted from comparative experiments. The detailed procedure is outlined in Algorithm 2.

Algorithm 2 Pseudocode for BAIGNet

Require: x_p: prefire image
Require: y: text prompt for wildfire image generation
Require: τ_θ: text encoder for the text prompt
Require: m: mask generated from VCMGNet
Require: model: diffusion model for generating mask
Require: t: time step
Ensure: x_{syn}: postfire image
  function BAIGNET(x_p, y, m, model, τ_θ, t)
    $\text{embeds}_{text} \leftarrow \tau_\theta.\text{encoder}(y)$
    $l_t \leftarrow \text{model.initlatent}(\text{embeds}_{text}, t)$
    $l_{condition} \leftarrow \text{model.initlatent}(\tau_\theta.\text{encoder}(m))$
    $l_{background} \leftarrow \text{model.initlatent}(\tau_\theta.\text{encoder}(x_p))$
    $l \leftarrow l_t + l_{condition} + l_{background}$
    $\text{pred}_{sd} \leftarrow \text{model.unet}(l, t)$
    $\text{pred}_{ctrl} \leftarrow \text{model.unet}(l, t)$
    $\text{pred} \leftarrow \text{pred}_{sd} + \text{pred}_{ctrl}$
    if inpainting:
      $m_c \leftarrow 1 - m$
      $subject \leftarrow pred \times \text{model.init}(m)$
      $background \leftarrow \text{model.init}(x_p \times m_c)$
      $pred \leftarrow subject + addnoise(background)$
    $x_{\text{syn}} \leftarrow \text{model.decode}(\text{pred})$
    return x_{syn}
  end function

IV. Experimental Setting

A. Datasets

We evaluate on the FLOGA dataset [34], which contains Sentinel-2 and MODIS images covering large regions of Greece and numerous wildfire events from 2017 to 2021. For each event, both prefire and postfire images are available, as well as expert-labeled burned area polygons from the Greek Fire Service, plus sea masks, land-cover maps, and cloud masks.

In this study, we used the near-infrared (NIR), short-wave infrared 1 (SWIR1), and short-wave infrared 2 (SWIR2) bands from the Sentinel-2 images, due to the fact that these bands are most sensitive to high-temperature regions [35]. Meanwhile, we combined the above three bands and generated a false-color image with a spatial resolution of 20 m for experimental validation. Finally, we cropped both the remote sensing image and the corresponding masks into 512×512 pixel patches to construct the training and testing set.

B. Implementation Details

The experiments are implemented on an Ubuntu system equipped with two NVIDIA GeForce RTX 3090 GPUs to accelerate the training process. The pretrained SD model [36] is used in our Wildfire-Diff. We use the AdamW optimizer and the cosine scheduling strategy with a base learning rate of $1e^{-4}$. The input size is 512×512, and the training epoch is set to 50. The batch size is set to 4. For downstream CD tasks, all models adopted the AdamW optimizer with a batch size of 4, a learning rate of $3e^{-4}$, and a weight decay parameter of $1e^{-2}$.

C. Comparison Methods

The comparison of generation performance includes the vegetation-constrained wildfire mask generation and the wildfire image generation network. In the vegetation-constrained

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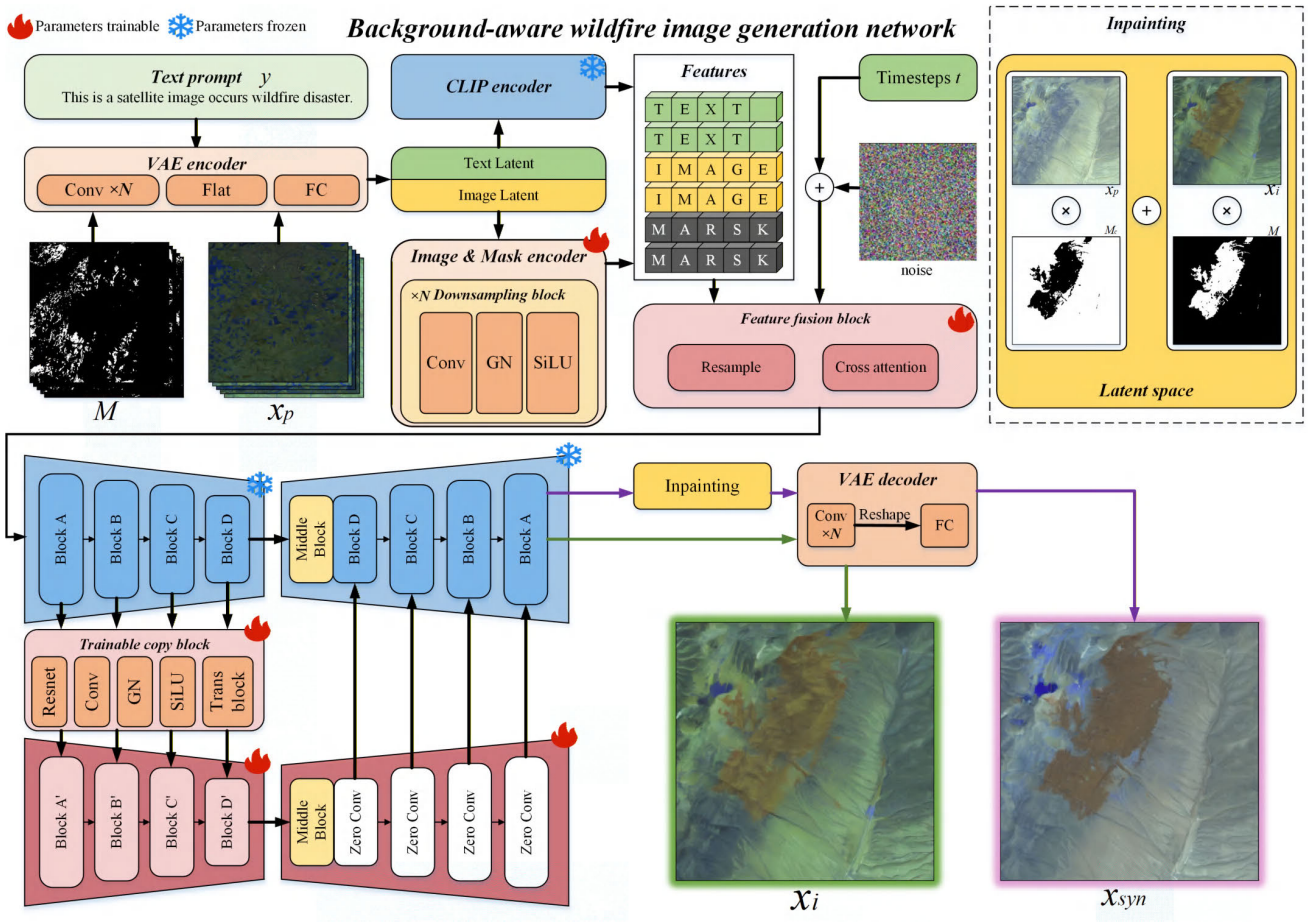


Fig. 4. Overview of BAIGNet. x_i is generated inpainting wildfire result and x_{syn} is generated wildfire result.

wildfire mask generation experiment, we present burned area mask generation results from the burned area mask diffusion module alongside manually annotated ground-truth masks, evaluated through both visual comparisons and quantitative metrics. In the wildfire image generation experiment, we systematically assess the effectiveness of BAIGNet in wildfire image generation through three aspects: visual characteristics analysis, spectral feature examination, and quantitative metrics evaluation.

We further study generated data in three settings: 1) data augmentation; 2) scaling the amount of generated data; and 3) performance under data scarcity. For augmentation, we train four CD models: BIT [37], CGNet [38], EfficientCD [39], and BAMCD [34]. For scaling, we test whether increasing only synthetic samples improves performance. For data scarcity, we test whether synthetic samples generated from validation prefire images help.

To mitigate potential label noise in FLOGA, we curated an additional test set of manually annotated pre/postfire pairs from the same Sentinel-2 scenes but annotated independently. In addition, as the inpainting results may impair background generalizability in CD and degrade performance, we only include inpainting results in our ablation study, excluding them from the main comparative experiments. All metrics are computed at the pixel level and averaged over multiple runs to reduce stochastic variance.

Finally, we conduct an ablation to quantify the effects of BAIGNet and the inpainting module on overall generation quality.

D. Evaluation Metrics

We evaluate with self-reference (within real data) and cross-reference (synthetic versus real) protocols. Metrics include SSIM [40] for spatial similarity, spectral cosine similarity (SCS) for frequency-domain alignment, FID [41] for distributional distance, inception score (IS) [9] for sharpness and diversity, PSNR [42], and LPIPS for perceptual quality. Unless stated otherwise, we use 2000 images per evaluation for stable estimates

$$SCS = \frac{\sum_{i=1}^{N} |F(x_i)| \times |F(y_i)|}{\sqrt{\sum_{i=1}^{N} |F(x_i)|^2} \sqrt{\sum_{i=1}^{N} |F(y_i)|^2}} \tag{7}$$

where $F(x)$ is the Fourier transform, x_i is the image from x datasets, and y_i is other image from y datasets.

In addition, for CD, we report overall accuracy (OA), IoU, precision, recall, and $F1$-score. These metrics can be formally defined as follows:

$$OA = \frac{\sum_{k=1}^{K} TP_k + TN_k}{\sum_{k=1}^{K} TP_k + FP_k + TN_k + FN_k} \tag{8}$$

$$IoU = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k + FP_k + FN_k} \tag{9}$$

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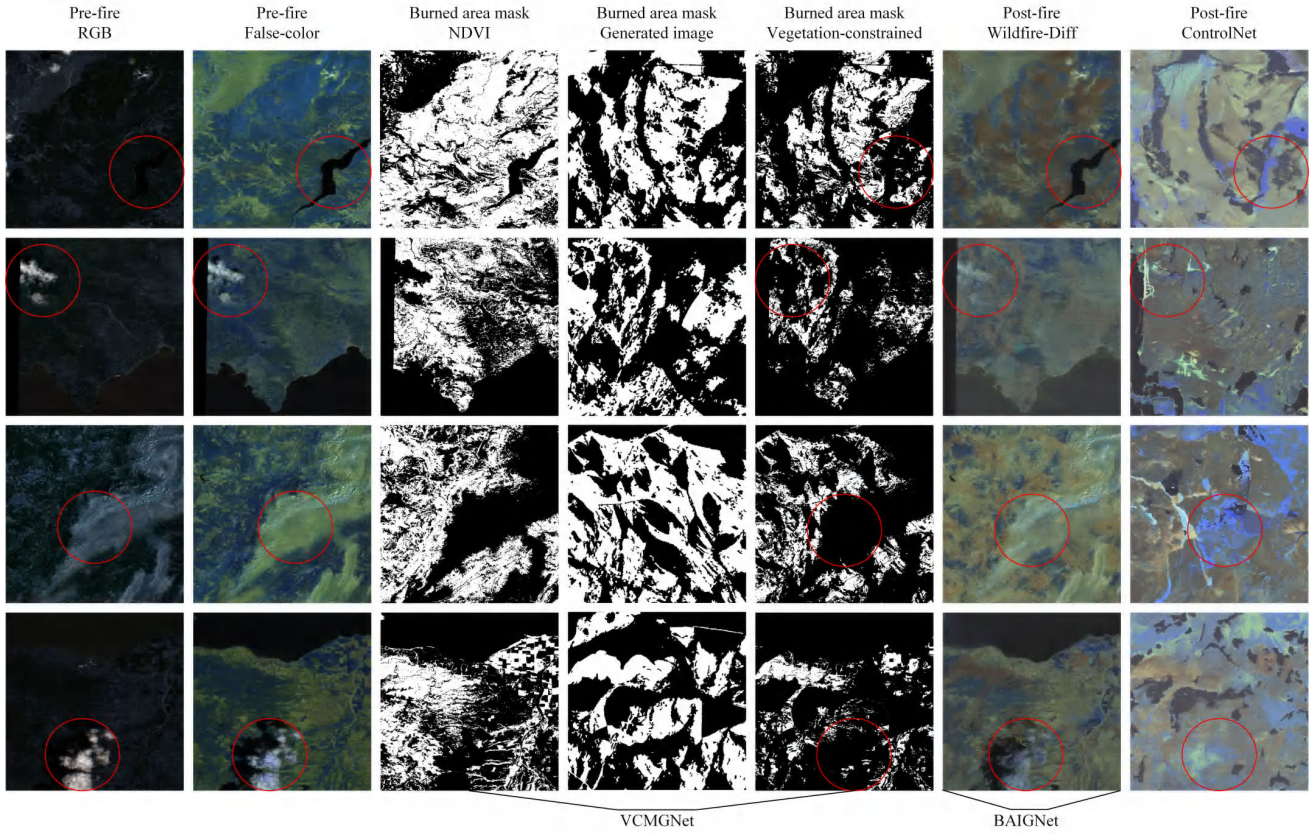


Fig. 5. Step-by-step visualization and comparison of the Wildfire-Diff generation pipeline.

$$precision = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{TP_k + FP_k} \tag{10}$$

$$recall = \frac{1}{K}\sum_{k=1}^{K}\frac{TP_k}{TP_k + FN_k} \tag{11}$$

$$F1 = 2 \times \frac{precision \times recall}{precision + recall} \tag{12}$$

where TP_k, FP_k, TN_k, and FN_k indicate the true positive, false positive, true negative, and false negatives, respectively, for the specific object indexed as class k. K is the number of categories. OA is computed for all categories, including the background pixels.

V. Experiments and Results

A. Comparison of Generation Performance

1) Performance of Vegetation-Constrained Wildfire Mask Generation: To evaluate the quality and similarity between the VCMGNet-generated masks and the ground truth masks, we evaluated using FID, IS, and LPIPS metrics to compare. As the results shown in Table I, the quantitative metrics for the proposed methods are slightly inferior to the ground-truth masks, with a deviation of around 29%, while still maintaining comparable performance levels. As a result, the generated masks are reliable enough to use as input masks in the following BAIGNet framework.

The stepwise procedure results of VCMGNet for burned area masks are shown in Fig. 5. Without the vegetation-constrained block, the results show slight deviations from the ground truth in spatial continuity and edge sharpness, though key morphological features are preserved. Incorporating the block eliminates these differences, yielding results that closely match the ground truth in both boundaries and morphology. The final burned area mask (shown circled in red in Fig. 5) preserves the features of nonburnable regions (e.g., rivers and bare soil) and retains the original image edges from dataset cropping. The resulting paired prefire and postfire images are of high quality and well-suited for subsequent wildfire analysis and extraction.

The VCMGNet module generates binary fire masks, controlled by different α parameters, as shown in Fig. 6. Three examples are presented: precise human-annotated masks, low-coverage masks with sparse fire regions, and high-coverage masks with dense fire regions. Visually, both generated types resemble the manual annotations in shape and distribution.

2) Comparisons of Wildfire Image Generation: To verify the effectiveness of the proposed framework, we compare Wildfire-Diff with: 1) SD v2.1 (pretrained on LAION-5B); 2) ControlNet fine-tuned on FLOGA; and 3) a copy–paste composite using FLOGA pre/postfire images. Real burned area masks are provided to guide generation for fair comparison against real postfire images.

a) Visual comparison: As highlighted in Fig. 5, BAIGNet uses prefire background features to preserve land cover (e.g., water, bare land, and clouds) outside the burned mask while generating fire-affected regions guided by VCMGNet.

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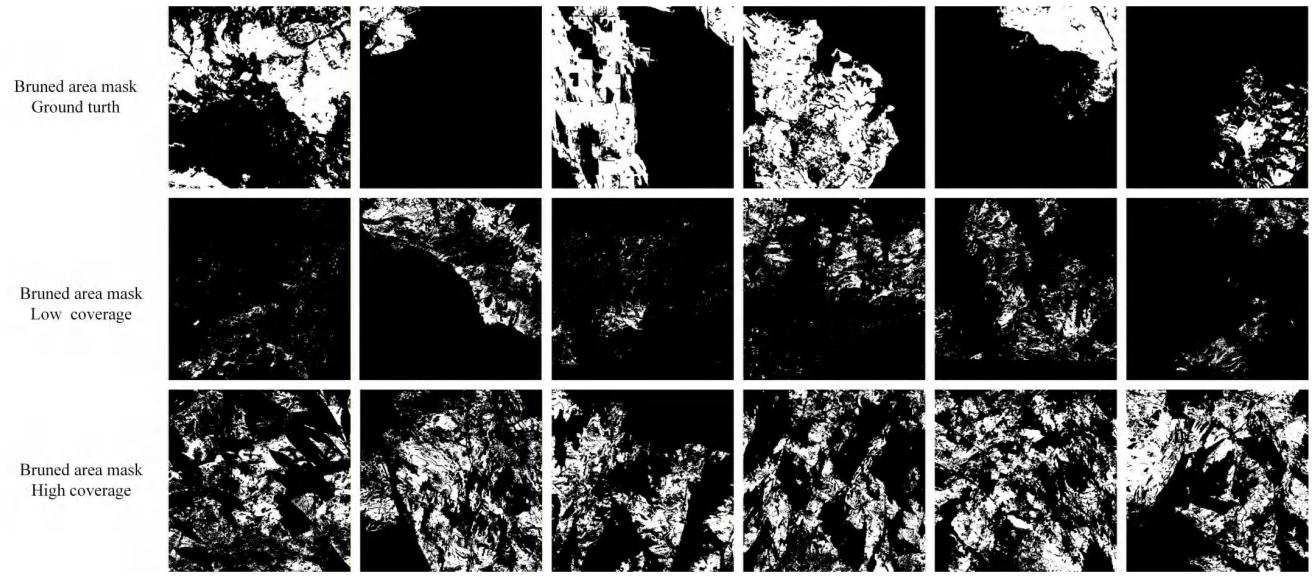


Fig. 6. Comparison of fire mask generation results.

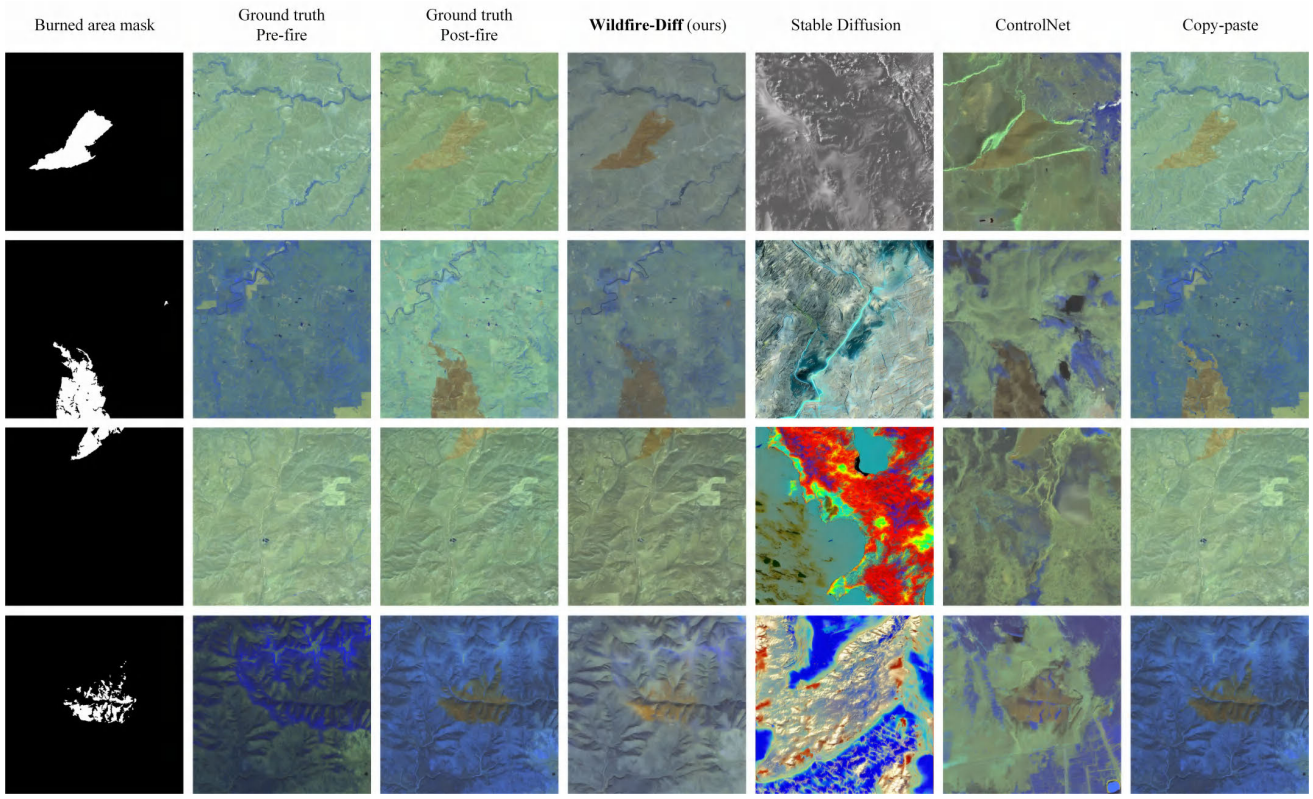


Fig. 7. Comparison of generation results for wildfire remote sensing images.

Fig. 7 shows results from all methods. SD exhibits limited capability in generating realistic wildfire infrared remote sensing images, largely due to its text–image pretraining framework, which fails to adequately represent the spectral characteristics of burned areas and backgrounds. Moreover, it struggles to effectively integrate auxiliary inputs, such as burned area masks and prefire images, resulting in substantial output variability. ControlNet produces wildfire images but exhibits critical flaws: 1) incoherent background landscapes mismatching real terrain; 2) blurred burned area boundaries hindering downstream tasks; and 3) spurious fire patterns beyond mask boundaries indicate poor constraint adher-

TABLE I
QUALITY EVALUATION OF BURNED AREA MASK GENERATION

Groups	FID↓	IS↑	LPIPS↓
Ground truth	**31.32**	**1.43**	**0.35**
Burned area mask diffusion	42.01	1.19	0.48

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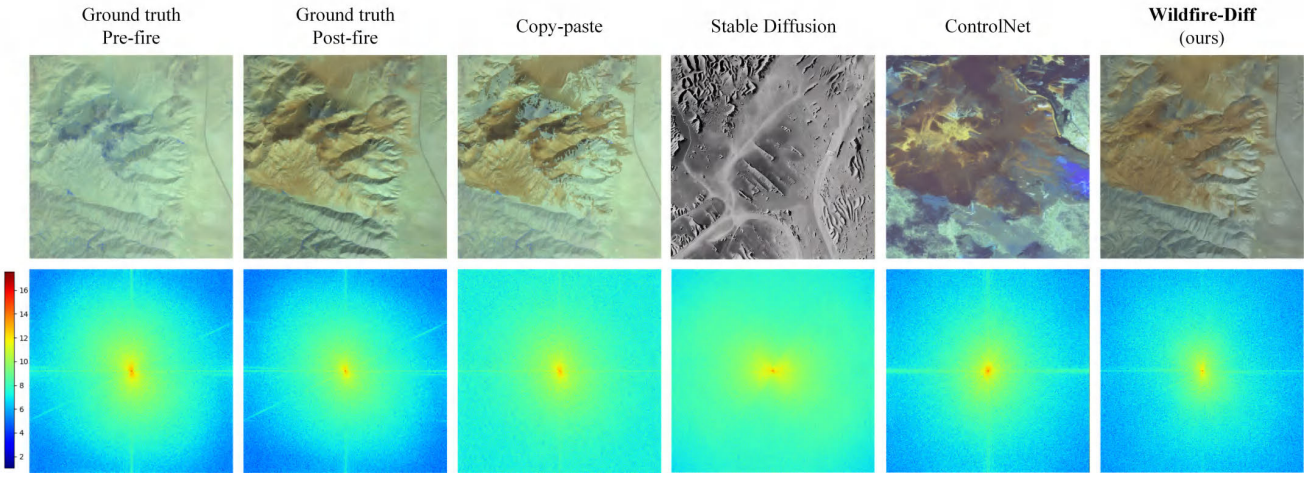


Fig. 8. Frequency-domain comparison of generation results for pseudocolor wildfire remote sensing images.

TABLE II

QUALITY EVALUATION OF GENERATION RESULTS FOR WILDFIRE REMOTE SENSING IMAGES

Benchmarks	Method	SSIM↑	SCS↑	FID↓	IS↑	PSNR↑	LPIPS↓
Real-Image	Self-Reference	1.00	1.00	0.00	1.07	inf	0.00
	Cross-Reference	0.39	0.98	16.96	1.08	14.32	0.67
	Copy-Paste	0.52	0.99	22.52	1.07	18.55	0.35
Synthetic-Image	Stable Diffusion	0.18	0.96	138.76	**1.15**	11.64	0.73
	ControlNet	0.34	0.98	40.34	1.06	14.31	0.67
	Wildfire-Diff	**0.59**	**0.99**	**30.42**	1.07	**17.21**	**0.46**

TABLE III

MODEL COMPLEXITY AND INFERENCE EFFICIENCY COMPARISON

Method	Base required	Params.(M)	Inference Time(s)
Stable Diffusion	-	1220.00	2.90
ControlNet	SD 2.1	364.23	4.01
Wildfire-Diff (VCMGNet)	SD 2.1	363.25	3.15
Wildfire-Diff (BAIGNet)	SD 2.1	365.32	3.78

TABLE IV

RESULTS OF TRAINING ON BIT

Percentage Blend (FLOGA - Synth)	Methods	OA	IoU	F1	Rec.	Pre.
100-0 (%)	-	93.66	32.86	49.46	45.29	54.48
0-100 (%) Synthetic-Only	Copy-Paste	89.30	35.65	52.56	**86.55**	37.74
	ControlNet	**93.24**	44.03	61.14	77.65	**50.41**
	Wildfire-Diff	92.80	**44.87**	**61.95**	85.59	48.54
100-50 (%) Hybrid A	Copy-Paste	92.23	41.79	58.95	81.49	46.17
	ControlNet	**95.38**	42.20	59.35	49.23	**74.71**
	Wildfire-Diff	92.80	**45.12**	**62.18**	**86.43**	48.56
100-100 (%) Hybrid B	Copy-Paste	92.20	41.58	58.74	**81.06**	46.06
	ControlNet	95.39	37.02	54.04	39.62	**84.98**
	Wildfire-Diff	**95.82**	**55.34**	**71.25**	75.73	67.27

ence. We construct synthetic samples by copying the burned regions from postfire images and pasting them onto prefire images, thereby simulating postfire scenes while preserving realistic optical characteristics. Although the composites may look plausible, edge discontinuities between pasted burned areas and the surrounding background expose spatial inconsistencies. These artifacts will be quantitatively analyzed via frequency-domain comparisons.

TABLE V

RESULTS OF TRAINING ON CGNET

Percentage Blend (FLOGA - Synth)	Methods	OA	IoU	F1	Rec.	Pre.
100-0 (%)	-	92.46	31.47	47.88	50.59	45.44
0-100 (%) Synthetic-only	Copy-Paste	86.26	32.24	48.76	**95.48**	32.74
	ControlNet	94.54	38.45	55.54	49.84	**62.71**
	Wildfire-Diff	**94.90**	**51.29**	**67.80**	78.50	59.67
100-50 (%) Hybrid-A	Copy-Paste	90.64	38.67	55.77	**86.23**	41.22
	ControlNet	**95.46**	**45.88**	**62.90**	56.19	**71.44**
	Wildfire-Diff	95.23	43.99	61.10	54.68	69.23
100-100 (%) Hybrid-B	Copy-Paste	90.04	37.68	54.73	**87.99**	39.72
	ControlNet	**95.49**	49.81	66.50	65.38	**67.65**
	Wildfire-Diff	94.93	**53.29**	**69.53**	84.54	59.05

TABLE VI

RESULTS OF TRAINING ON EFFICIENTCD

Percentage Blend (FLOGA - Synth)	Methods	OA	IoU	F1	Rec.	Pre.
100-0 (%)	-	94.25	13.00	23.01	19.06	29.04
0-100 (%) Synthetic-only	Copy-Paste	90.90	25.24	40.31	68.14	28.62
	ControlNet	**94.56**	21.65	35.59	33.35	38.16
	Wildfire-Diff	92.14	**40.68**	**57.84**	**78.76**	**45.70**
100-50 (%) Hybrid-A	Copy-Paste	91.61	25.86	41.09	**64.86**	30.07
	ControlNet	**95.25**	18.99	31.92	24.68	**45.17**
	Wildfire-Diff	93.89	**26.40**	**41.77**	48.61	36.61
100-100 (%) Hybrid-B	Copy-Paste	94.73	21.77	35.76	32.49	39.76
	ControlNet	**95.30**	20.88	34.55	27.49	46.50
	Wildfire-Diff	93.66	**39.43**	**56.56**	**60.29**	**53.26**

b) Image spectral fidelity comparison and evaluation: As illustrated in Fig. 8, frequency-domain differences between generated results and real images were analyzed to assess

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TABLE VII
RESULTS OF TRAINING ON BAMCD

Percentage Blend (FLOGA : Synth)	Methods	OA	IoU	F1	Rec.	Pre.
100:0	-	91.48	30.40	46.62	54.33	40.83
0:100	Copy-Paste	64.72	15.93	27.49	**97.69**	15.99
	Controlnet	91.18	33.69	50.40	90.08	34.98
	Ours	**95.75**	**57.60**	**73.10**	85.52	**63.82**
100:50	Copy-Paste	76.00	16.85	28.84	**96.53**	16.95
	Controlnet	**96.73**	49.75	66.44	54.90	**84.13**
	Ours	96.22	**60.45**	**75.35**	80.34	70.94
100:100	Copy-Paste	83.11	22.15	36.27	**96.87**	22.31
	Controlnet	95.96	53.70	69.88	61.46	**80.97**
	Ours	**96.91**	**67.21**	**80.39**	82.48	78.40

TABLE VIII
RESULTS OF THE DATA SCALING EXPERIMENT FOR GENERATED SAMPLES

Sample size multiple	OA	IoU	F1	Rec.	Pre.
1×	92.08	47.81	64.70	**86.73**	51.50
2×	94.20	50.82	67.40	84.83	55.91
4×	**95.00**	52.78	69.09	81.16	**60.15**
8×	94.48	**53.06**	**69.33**	82.95	59.55

TABLE IX
RESULTS OF THE DATA SCARCITY EXPERIMENT FOR WILDFIRE-DIFF

Methods	Categories	OA	IoU	F1	Rec.	Pre.
BIT	Original	84.04	38.24	55.33	**83.95**	41.26
	Enhance	**93.64**	**58.37**	**73.72**	75.73	**71.80**
CGNet	Original	84.81	38.76	55.86	81.64	42.46
	Enhance	**93.41**	**62.19**	**76.69**	**92.09**	**65.70**
EfficientCD	Original	91.31	39.49	56.62	48.18	68.63
	Enhance	**91.91**	**42.85**	**59.99**	**51.49**	**71.85**
BAMCD	Original	91.17	37.72	54.78	39.91	87.32
	Enhance	**91.96**	**41.22**	**58.37**	**43.75**	**87.70**

spectral fidelity. Both copy–paste and SD methods exhibit amplified high-frequency components relative to real images, leading to noticeable discrepancies in edge details and noise patterns. Although ControlNet aligns more closely with the spectral characteristics of real images, it introduces excessive smoothing in mid-frequency regions. Despite minor high-frequency deviations, Wildfire-Diff achieves the closest overall spectral match to real postfire images, excelling in both low-frequency patterns and broader spectral fidelity. These observations will be quantified using SCS metrics.

c) Comparison of generation quality: Table II summarizes five metrics over two groups (real-image references and synthetic-image baselines). Within the synthetic-image group, Wildfire-Diff leads on all metrics except IS. ControlNet and SD trail due to higher generation randomness. Notably, Wildfire-Diff attained an SCS score of 0.99, matching the performance of the copy–paste method. This score (where a value of 1.0 indicates identical images) demonstrates that its spectral features closely resemble those of real remote sensing images, providing a strong foundation for enhancing datasets in downstream tasks. An FID score of 30.42 further confirms the close alignment between the distributions of synthetic and real data. Interestingly, the SD achieved the highest IS score, likely due to its inherent randomness, enhancing dataset variety and resolution. The 20-m spatial resolution limitation inherent to Sentinel-2-based wildfire images reduces visual clarity compared to the SD outputs. Importantly, the IS scores of Wildfire-Diff aligned well with the synthetic-image benchmarks, indicating its outputs effectively replicate the features of real data.

In addition to generation quality, we evaluated the computational complexity and inference efficiency of the proposed model on a single NVIDIA RTX 3090 GPU. As shown in Table III, SD requires 2.9 s per inference; ControlNet (SD 2.1) has 364.2M parameters and an inference time of 4.01 s, and Wildfire-Diff in the VCMGNet and BAIGNet variants has 363.3M and 365.3M parameters, with inference times of 3.15 and 3.78 s, respectively, and the full pipeline completes in 6.93 s. These results indicate that Wildfire-Diff achieves efficient generation while maintaining comparable model complexity.

B. Experiments for Generated Data

1) Experiments on Data Augmentation: To evaluate the impact of generated datasets on downstream tasks, we evaluate three synthetic datasets (Wildfire-Diff, copy–paste, and ControlNet) with four CD models (BIT, CGNet, EfficientCD, and BAMCD), focusing on metrics IoU and $F1$. Four training regimes are used: 1) Baseline: 100% FLOGA; 2) Synthetic-only: 100% synthetic; 3) Hybrid A: 100% FLOGA + 50% synthetic; and 4) Hybrid B: 100% FLOGA + 100% synthetic.

Wildfire-Diff demonstrated clear advantages in the experiments (see Tables IV–VII). In the synthetic-only group, it achieved the highest IoU and $F1$ scores, surpassing the second-best method by 44.46% and 25.41%, respectively. By contrast, the copy–paste strategy underperformed relative to the baseline when combined with BIT and CGNet. Similar results were observed in the Hybrid-B group, where Wildfire-Diff exceeded the next-best method by 36.58% in IoU and 24.76% in $F1$.

Data augmentation experiments revealed a consistent performance improvement as Wildfire-Diff-generated data was added to the FLOGA dataset in 50% increments. During the enhancement phase, Wildfire-Diff achieved significantly higher average relative IoU gains (92.64%) compared to copy–paste (23.87%) and ControlNet (49.01%). These results confirm the superior performance of Wildfire-Diff for data augmentation and underscore the synergistic effect of combining synthetic and real data. Specifically, Wildfire-Diff effectively complements the distribution of real samples by filling critical gaps in pattern coverage.

Notably, the performance of ControlNet occasionally matches Wildfire-Diff for CGNet in terms of IoU, because both are fine-tuned diffusion models whose latent features capture realistic burn textures and spectral cues. CGNet also benefits from diverse backgrounds produced by diffusion, improving robustness under high background variability. In addition, wildfire CD differs from building CD: fires spread rapidly, so temporal gaps are short. Wildfire-Diff, therefore, preserves prefire backgrounds while applying controlled style changes to simulate realistic temporal differences. Even when ControlNet matches IoU in some cases, Wildfire-Diff consistently delivers higher generation quality and downstream accuracy.

2) Scaling Experiments of Generated Samples: To evaluate the effect of scaling generated data on downstream task

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TABLE X
RESULTS OF THE MODULAR ABLATION EXPERIMENT FOR WILDFIRE-DIFF

Model	BAIGNet	Inpainting	SSIM↑	SCS↑	FID↓	IS↑	PSNR↑	LPIPS↓
Wildfire-Diff	×	×	0.34	0.98	40.34	1.06	14.31	0.67
	✓	×	<u>0.55</u>	**0.99**	<u>31.67</u>	**1.07**	<u>16.82</u>	**0.45**
	×	✓	0.43	0.98	43.89	1.06	15.74	0.61
	✓	✓	**0.59**	**0.99**	**30.43**	**1.07**	**17.21**	**0.45**

performance, we study scale by synthesizing 1×, 2×, 4×, and 8× the baseline number of samples with Wildfire-Diff (baseline = original FLOGA size). BAMCD, chosen from preliminary tests, is trained under identical settings on each set (see Table VIII).

The results demonstrate that the doubling of the number of samples generated by Wildfire-Diff yielded average gains of 3.55% in IoU and 2.34% in $F1$. As the number of samples kept growing, the performance gains showed a slight slowdown. This suggests that the fit of the model to the data gradually approaches saturation. Although the recall of the 1× setting is slightly higher than that of 8×, this is mainly because the proportion of positive samples in the wildfire dataset is naturally much smaller than that of negative samples. As the dataset size increases, the imbalance between positive and negative samples becomes more pronounced, causing the model to be biased toward predicting negative samples. In addition, with more training data, the model tends to adopt a more conservative decision strategy to reduce false positives, which further lowers recall while improving overall precision and stability.

Overall, this data scaling experiment confirms that expanding the quantity of samples generated by Wildfire-Diff can significantly enhance CD model performance. It also reveals the pattern of performance growth when leveraging large-scale synthetic data.

3) Performance of Sample-Scarce Scenarios: To evaluate whether Wildfire-Diff-generated data enhances the performance of downstream models under data scarcity, we sample 1000 FLOGA instances and split them 1:9 (train:test). From test-set prefire images, Wildfire-Diff generates 900 postfire images with distinct masks. We form two training sets: original (100 real pairs) and enhanced (100 real + 900 synthetic), and evaluate BIT, CGNet, and EfficientCD on the same test set (see Table IX).

Results indicate that training solely on limited real data yields suboptimal performance, as expected. However, incorporating the generated data enhanced the performance greatly. Specifically, models exhibited a mean increase of 32.71% in IoU and 20.75% in $F1$. This confirms that Wildfire-Diff-generated data, derived from real prefire images, significantly enhances downstream task performance under data-scarce conditions.

C. Ablation Study

To compare the impact of different components within the proposed Wildfire-Diff framework, we replaced BAIGNet with ControlNet in the Wildfire-Diff framework, thereby removing background feature utilization. In addition, we introduced the inpainting block to assess its contribution. Model performance was evaluated using image quality metrics (see Table X).

Experimental results demonstrate that the configuration combining both BAIGNet and inpainting modules yields generated data of the highest quality, most closely approximating real observations. Utilizing BAIGNet alone provides the next-best performance. Critically, the BAIGNet module delivers the most significant performance gain, achieving an average improvement of 19.65% over the baseline. Furthermore, integrating the inpainting module provides an additional average quality improvement of 4.28% compared to using BAIGNet only. This enhancement is primarily attributed to the strategic incorporation of real prefire data by the inpainting module during generation. These findings confirm the significant individual contributions of both modules to wildfire image generation quality and highlight their generation effect, which collectively enables Wildfire-Diff to achieve state-of-the-art performance.

VI. CONCLUSION

In this article, we proposed a novel two-stage diffusion-based framework (Wildfire-Diff) for generating paired wildfire images and masks, aiming to address the scarcity of training samples in burned area mapping. In the first stage of our Wildfire-Diff, the proposed VCMGNet utilizes vegetation index constraints to produce burned area masks that not only exhibit realistic wildfire morphology but also have spatially plausible locations. In the second stage, the proposed BAIGNet generates realistic postfire images by utilizing prefire background features.

The experimental results demonstrate that our Wildfire-Diff outperforms several recent image generation methods in producing high-quality and trainable remote sensing wildfire images. Moreover, the proposed Wildfire-Diff can provide a flexible and effective data augmentation solution to mitigate the lack of training data in wildfire mapping, promoting research in this field. In future work, we will extend our Wildfire-Diff to other downstream applications and investigate the integration of multimodal learning to further improve its adaptability and generalization.

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